

MA 654: Financial Engineering

Course Code	MA 654
Course Name	Financial Engineering
L-T-P-C	2-1-0-3
Prerequisites	Good knowledge of multivariable calculus, probability, statistics and random process and at least one of C/C++/MATLAB/R/Spread sheets Packages.
Intended for	Ph.D./ M.S./B.Tech
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)
Replaced by	MA 545 by the 65th BoA (Aug 2026)
Discontinued	65th BoA (Aug 2026)

Course Contents

- **Module I**
Some Basic Definitions and Terminology. [2 Lectures]
- **Module II**
Forward and Future contracts, Definition and pricing, dividends and transaction costs. [3 Lectures]
- **Module III**
Efficient Market Hypothesis, Discrete and continuous random variable, Concept of Arbitrage, Duality theorem in LP and Fundamental theorem. [3 Lectures]
- **Module IV**
Asset Price Moment, Introduction to Option Markets, Options Valuations, Basic Option Theory, The Binomial Model, Black-Scholes Model and solution, Time dependency and change of variable, The Greeks and their properties. [10 Lectures]
- **Module V**
Bullish, Bearish and Neutral options strategies. [6 Lectures]
- **Module VI**
General principle of Hedging, Different types of Hedging, Delta Hedging, Delta Neutral Portfolio, Gamma Neutral Portfolio. [6 Lectures]

Text Books

1. J.C. Hull, Options, **Futures and Other Derivatives**, 7th Edition, Prentice Hall of India, 2006.
2. M. Capinski and T. Zastawniak, **Mathematics for Finance: An Introduction to Financial Engineering**, Springer, 2005.
3. S. Roman, **Introduction to the Mathematics of Finance: From Risk Management to Options Pricing**, Springer, 2004.

References

1. N. H. Bingham and R. Kiesel, **Risk Neutral Valuation**, 2nd Edition, Springer, 2004.
2. Simon Benninga, **Financial Modeling**, 3rd edition, MIT Press, 2008.
3. Ralf Korn, Elke Korn, **Option Pricing and Portfolio Optimization**, American Mathematical Society, 2000.